

Neeraj Kumar Pola

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Professional Summary

Built a production-grade RAG system handling 1,000+ daily queries at 1 to 1.5 ms latency. Specialize in agentic AI pipelines, LLM systems, and NLP engineering, deployed on Azure and AWS. Track record of cutting operational latency and improving model accuracy, including a 12× processing speedup at Media Sales Plus. Currently leading company-wide AI automation at VRT Management using LangChain and n8n.

Professional Experience

AI Engineer, VRT Management, LLC

Mar 2026 – Present

- Led a company-wide AI automation initiative using **n8n** and **LangChain**, automating 8+ business workflows and cutting manual processing time by 70%.
- Built an 8-agent recruitment pipeline (**LangGraph**, **FastAPI**, Streamlit) that turns a raw job description into a ranked, audited shortlist with personalized outreach in under 2 minutes for \$0.05 in LLM cost, validated at a 100% pass rate across a 10-job regression suite.
- Built a **hybrid retrieval** and scoring system (vector search, keyword search, cross-encoder reranking) running 300+ evidence-backed evaluations per job, with bias safeguards that screen job descriptions in under 2 seconds and strip names before scoring.
- Simplified candidate sourcing by replacing three separate job board integrations with a single spreadsheet upload, cutting integration overhead while holding 100% recall on duplicate detection across 110 candidates.

Machine Learning Engineer, Media Sales Plus

Aug 2025 – Dec 2025

- Built and deployed a production **NLP system** as a Flask REST API on Azure App Service, serving internal editorial workflows with sub-200 ms inference latency.
- Implemented end-to-end ML data pipelines with automated validation, profanity detection, and error handling. Raised processing accuracy from 95% to 98% and cut processing time from 60 minutes to 5 minutes, a **12× speedup**.
- Designed **spaCy NER** and rule-based weak supervision workflows with entity normalization and full correction traceability, enabling scalable downstream analytics.

Artificial Intelligence Intern, CAIR, DRDO

Oct 2023 – Mar 2024

- Fine-tuned multilingual **Wav2Vec2** models (HuggingFace Transformers) on 45+ hours of Bengali and Assamese speech data, achieving 12–18% improvement in recognition accuracy, and engineered a custom 512-unit neural layer for BPE transfer learning on 27 hours of curated low-resource audio.
- Built modular preprocessing pipelines (audio segmentation, tokenization, alignment) and maintained training/evaluation infrastructure tracking WER and CER metrics.

Machine Learning Intern, Feynn Labs

Mar 2023 – May 2023

- Built time-series forecasting models using SARIMA, ARIMA, and Prophet on 10,000+ records with feature engineering (seasonality, lags, trends), improving prediction accuracy by 15%; created Tableau and Power BI dashboards for stakeholder KPI reporting.

Projects

PaperBuddy – Multi-Agent Academic Paper Analysis

Personal Research

- Designed and built a 12-agent **LangGraph** system combining parallel specialist agents, a critic-driven reflection loop, and an orchestrator for targeted revisions, replacing manual triage with full paper analysis in 20 to 30 seconds at \$0.0018 per paper.
- Built an evaluation harness (BERTScore, LLM-as-judge, precision/recall) and ground-truthed 10 papers across NLP, CV, and RL domains, reaching 86.2% claim-extraction F1.
- Implemented a multi-turn Q&A agent with 6 tool-calling functions over a hybrid retrieval pipeline (BM25, embeddings, reciprocal rank fusion) and shipped a Streamlit interface with an interactive graph visualization, turning the CLI pipeline into a demoable product.

Automatic Speech Recognition (ASR) for Low-Resource Languages

Personal Research

- Benchmarked three ASR architectures (**Wav2Vec2**, **Wav2Vec**, **Whisper**) on Bengali and Assamese speech data, evaluating WER and CER across model sizes to identify the best architecture for each resource constraint.
- Compared three adaptation strategies (fine-tuning, transfer learning, BPE tokenizer adaptation) with custom data augmentation. Transfer learning with BPE adaptation achieved the lowest WER on the most resource-scarce subsets, with all pipelines and results published as reproducible scripts on GitHub.

Comparative Reinforcement Learning Benchmarking

University Project

- Implemented and benchmarked 6 RL algorithms (**PPO**, **A2C**, **DQN**, **DDQN**, **TD3**, **DDPG**) in PyTorch and OpenAI Gym across 5 environments spanning discrete (CartPole, LunarLander, Atari) and continuous (MountainCar, Pendulum) action spaces.
- Ran hyperparameter sensitivity and convergence analysis across all 6 algorithms using standardized evaluation protocols, finding PPO more sample efficient than A2C in discrete spaces and TD3 more stable than DDPG in continuous domains; published the full training infrastructure and results on GitHub.

Education

University at Buffalo, M.S. in Artificial Intelligence GPA: 3.46/4

Aug 2024 – Dec 2025

- Coursework: Machine Learning, Deep Learning, Data Intensive Computing, Reinforcement Learning, Algorithms

Technical Skills

AI & Machine Learning: PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers, NLP, spaCy, Time Series Analysis, LLMs, RAG Systems, Agentic AI, LangChain, LangGraph, LangSmith, N8N

Data & Analytics: Pandas, NumPy, Spark, Airflow, ETL Pipelines, Feature Engineering, Tableau, Power BI, Matplotlib

Programming: Python, SQL, JavaScript, Java, C, R

Cloud & Databases: Azure App Service, AWS, GCP, Snowflake, MongoDB, MySQL, TiDB, FAISS, ChromaDB

APIs & Backend: FastAPI, Flask, REST APIs, React, Docker, CI/CD

Certifications: AWS Cloud Practitioner, Snowflake Data Engineering, Snowflake Data Science.